

RESEARCH ARTICLE

Ransomware Attacks in Cyber-Physical Systems: Countermeasure of Attack Vectors Through Automated Web Defenses

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ABSTRACT Ransomware has appeared to be the most damaging and devastating type of malware attack in any cyber physical system. The resilience of a web browser to deal with the malware attack is of significance importance, however, evaluating the performance of a browser to tackle these attacks is a challenging task. Due to various automation techniques, web applications can be tested without human intervention. Technologies such as Junit, Chakram, and Selenium are useful in automated testing but the problem is that the attacker uses harmful code and automated web approaches to distribute their malware. In this research, our contribution is twofold. Firstly, we examine a new attack vector that cyber adversaries can possibly use in the future to infect an operating system with a malware. Currently, attackers use various techniques to gain access to victims' personal computers. Secondly, we present a novel automated web defence to countermeasure these malware attacks. The proposed research aims to provide a better understanding of the new computer virus-spreading techniques that intruders can use in the future. We provide the insight of these attacks and present ways to countermeasure the attacks and to reduce the attack surface. Experiments and flow diagrams have been used to demonstrate the attack and defence approach. To offer malware lateral movement and to encrypt the data of users' device, we use Selenium automation tool on a social media platform. For our experimentation, we developed an application which has been tested on a variety of browsers including Google Chrome, Firefox, and Safari. Our research has revealed that we have an 85 percent success rate when testing in a head-on environment. We have expanded our experiments on headless applications and interestingly, the accuracy rate improved and the probability of success increased to 95 percent. Lastly, we have demonstrated a unique method for detecting and stopping web automation that is generally applicable.

INDEX TERMS Ransomware, selenium web automation, IoT, social sites, offensive security.

I. INTRODUCTION

Ransom belongs to a malware family that can obfuscate and lock computer data on a victim's machine. Typically, Malware is a special type of virus that is concealed in another

program to delete, modify, or encrypt files. The malware has the capability to launch a harmful, invasive application that damages the protection, safety, and functionality of the computer of the client [1]. Malware is the most destructive threat to any type of host machine and causes devastation and instability. Corporations and enterprise networks are susceptible to cyber attacks from several classes of malware

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families. The attackers then demand a ransom from the user to regain access to their information upon payment. Their range of ransomware vectors will necessitate access to a laptop. One of the most common delivery systems is phishing spam attachments that return to the victim in an email, masquerading as a file they must trust [2]. There are several things the malware might do once it has taken over the computer of a victim. Ransomware is malware that uses an exploitation strategy similar to cipher attacks to block user folders and syndicated data, then demands cryptocurrency in exchange for secure transaction. Some ransomware gets into the system utilizing social engineering, malicious advertisements, spamming, and drive-by downloads, whereas others try to discover vulnerabilities to take advantage of them. exploitable via a communication endpoint or a vulnerable “door-to-door” attack [3].

It is important to be aware of conquerable testing and security loopholes, as well as the various forms of exploiting ransomware as a service. Some notable attacks in the past include Habana Labs and Randstad in December 2020, Bad Rabbit in 2017, and Ryuk ransomware in 2018 [4]. Figure 1 outlines a basic technique used by attackers to insert malware on the victim’s machine and obtain their credentials. This technique involves paying for malicious application cipher codes, without requiring any coding skills. The value of these codes may be determined through explicit purchase or the use of a ratio theme, indicating the existence of a collaborative effort among cyber criminals.

It is important to analyze and identify the methods used by attackers as ransomware attacks may continue to occur in the future. In the Gregorian calendar month of 2020, “one in four attacks that IBM Security X-Force Incident Response has remediated this year are caused by ransomware” [5]. These incidents of malware appeared to be increasing in June 2020, accounting for a significant portion of all cyber-attacks remediated by IBM Safety X Force [5]. For example, the Habana Labs attack on December 13 involved a worm that creates AI systems and can withstand a programmatic weapon such as the Pay-2-Key virus. Figure 2 illustrates the attack vectors observed in February 2021, while Figure 3 shows the increasing trend of ransomware attacks between 2015 and 2019 [6]. This attack resulted in stolen sensitive information such as code and various business documents that were later published online [6]. Such attacks on Israeli-based corporations have become a growing trend in 2020 [7]. Check-Point’s research report states that Pay2Key is the most recent targeted ransomware attack on Israeli organizations with a focus on Iran. Randstad, one of the largest staffing organizations and owner of www.monster.com, was also a recent victim of an attack utilizing the Egregor ransomware [4].

Though it appears that the crackers have not demanded a payment quantity, the corporation acknowledges that the hackers have revealed important information [8]. The incident’s subsequent official dispatch was issued by the Netherlands

organization. They informed us that the investigation was complete and that it was discovered that the Egregor malware breached our IT systems and harvested our organization’s critical information, which included information related to our company’s operations in North America, Europe, Poland, and France. They must now print what is claimed to be a collection of that information [9]. The investigation is currently focused on spotting what information has been accessed, together with personal information, so that we can take applicable action with relevant characteristics and methods known as a “drive-by attack” wherever insecure websites are targeted. In a drive-by ransomware attack, a user visits a legitimate website, not knowing that their computer has been compromised by a hacker. Ryuk-Ryuk: That computer virus, shown in August 2018, paralysed the operating system’s reinstall option, making it impossible to revive locked files without a backup. Ryuk also encrypted network drives [10]. The results were devastating, and many US-based organizations paid the demanded ransoms in August 2018, according to reports, which estimated that over 640,000 dollars in ransom money was raised. Malware operation procedure: the primary way of becoming a victim of ransomware is by the user intervening. Malicious software is necessary for the methodology since the user must click on a suspicious link or transfer a malicious file (or files). On most occasions, it comes as an associated email. A download is applied when the link is clicked. or the attachment is downloaded. This transferor data then communicates with the domains or hosts that are controlled by the evil door behind the attack and asks the server to download the infected files. Figure 2 shows the prediction growth of cyber-attack loss [11]. Similarly, Ozer and Bilal proposed a center named “Global Ransom” that saves the maximum amount of data to create adequate strategies for cyber-attacks and find the person behind these exploits [12]. Although the FBI plays a central role in finding these attacks, they keep the information private, so this study provides access for researchers to work with law enforcement agencies.

The malware program is sent by the command-and-control servers. The vulnerability may arise from an unguarded feature within the software that operates similarly to the malware. Ransomware is a type of cyber-attack that typically involves encrypting the victim’s data using a strong encryption algorithm and demanding a ransom (usually in the form of Bitcoins) for decrypting the key. An example of such an attack is “Bad Rabbit,” which occurred in 2017 and was named after the two-word combination “Ransom and Ware” to denote the combination of money and goods. The report in Figure 3 shows the monthly collection of Varonis Forensics Team records that were actively detected while investigating incidents and reverse-engineering malware samples. This report is intended to provide a better understanding of the evolving cyber threat landscape and assist in adopting protective measures. This paper provides a detailed analysis of machine-virus

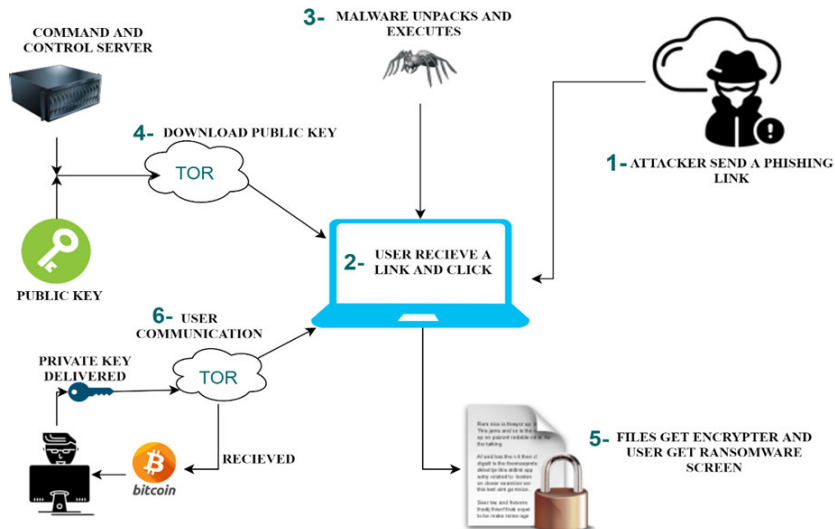


FIGURE 1. How ransomware works.

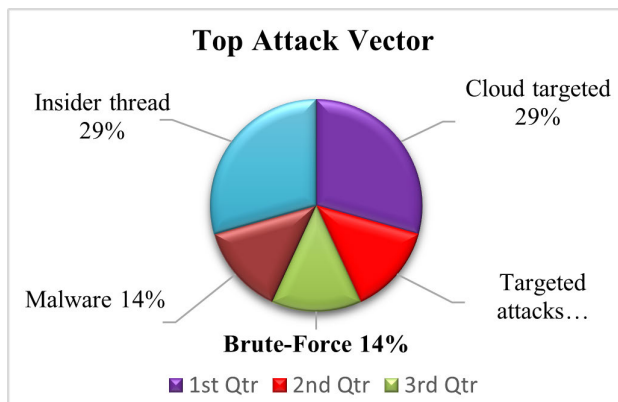


FIGURE 2. Attack vectors observed in February 2021.

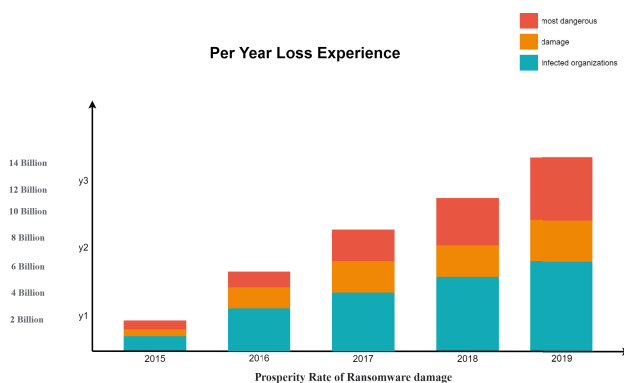


FIGURE 3. Year wise widening of ransomware.

recovery, mitigation, and prevention approaches for future research and analysis. Additionally, knowledge of viruses aids programmers and software developers in their efforts to find adequate solutions [14]. While the data collected ideally might also serve as a starting point for creating and improving

simpler ransomware protection systems, the rest of the case study is organised as follows [16].

A. OVERVIEW OF RESEARCH RELATED TO THE PHYSICAL LAYER

The study of cooperative fault-tolerant output regulation for linear heterogeneous multi-agent systems focuses on guaranteeing that a network of different and linked agents can maintain reliable and stable performance in the presence of defects. The study proposes an adaptive dynamic event-triggered system to handle this regulation. This method dynamically modifies the frequency and circumstances under which interactions and updates occur in response to the system's current state and behavior. This strategy allows the system to efficiently manage errors and disturbances while reducing needless communication and processing overhead. The adaptive trigger mechanism enables the system to respond efficiently to changing conditions, improving overall resilience and fault tolerance in multi-agent networks [59]. This study investigates cooperative output control of multi-agent systems (MASs) using a unique parameter convergence condition and prescribed-time event-triggered observers. The technique seeks to coordinate and regulate the outputs of numerous agents while maintaining precise control and stability in dynamic situations. The suggested technique offers a new parameter convergence condition, allowing for more effective agent coordination. Furthermore, prescribed-time event-triggered observers are used to efficiently handle communication and updates, ensuring that the system's reaction is prompt and exact. This framework is used to robot formation situations, proving its capacity to produce desired forms while maintaining robot coordination in the face of fluctuations and disruptions. The technique improves performance and resource efficiency while solving critical difficulties in cooperative multi-agent control [60].

B. OBJECTIVE

The objective of the paper is to use a novel method in which the Selenium Web Automation Tool is used to let ransomware spread laterally on social networks. Firstly, we demonstrate techniques for decrypting ransomware in this paper. Secondly, automation techniques are used for the lateral movement of the malware. We test our exploits on several browsers and get results in the form of graphs. We provide evidences and empirical studies to show that why ransomware attacks are successful. The rest of the paper is organized as follow: Section II reviews related studies concerning detection and mitigation techniques, and the author demonstrates bug families in general. The details of the datasets used in these analyses and taxonomy diagrams, as well as details that can help the security personnel to identify the ransom attacks are well elaborated in this section. In Section III, the design methodology of our proposed research is presented. In Section IV, we demonstrate the details of experiments that have been conducted and we analyze the outcomes. In Section V, the future trends are illustrated in details. Section VI discusses the idea of the proposed solution and in Section VII the paper is concluded.

II. LITERATURE REVIEW

In [1], the authors explained the effects of malware on the Windows environment. As we all know, the Windows platform is widely used, and the monitoring of viruses is done based on the behavior of the malware families. Securing the Windows environment is possible by keeping an eye on the system files and registry data. They experimented by using Oracle VM VirtualBox, Windows 7 and 10, and Cuckoo Sandbox. They conclude that protecting the system against malicious programs is achievable by analyzing the Windows registry and PC files. In their paper, Xueqin Zhang and Shinan Zhu present TGAN-IDS, a method for detecting encrypted ransomware attacks that is based on dual generative adversarial networks. This method is used to find ransomware attacks that haven't been seen before. In TGANIDS, DCGAN is mostly used to improve a generator that can make strong pseudo samples. Experiments show that the TGAN-IDS can make it much easier for an anomaly detector to spot a ransomware attack that isn't encrypted [25]. The author Junwei, Ruixuan, and Yu Jiang in their paper propose MGOPDroid, an effective anti-obfuscation Android malware detection technology. To address the issue that the malicious obfuscation version produced by the Android application obfuscation technology evades the current detection techniques, they offer MGOPDroid. For identifying Android malware obfuscation variations, MGOPDroid develops a unique feature selection technique and classification model. Then it performs Android malware detection based on multi-granular opcode characteristics [4].

In their paper [51], Bekkers et al. investigate the motivators behind entrepreneurs taking protective measures against ransomware attacks in their businesses. Using an extended

protection motivation theory model, the study examines the psychological factors that influence entrepreneurs' decisions to take preventive measures against cyber attacks. The research results indicate that perceived vulnerability, response efficacy, and self-efficacy play significant roles in entrepreneurs' intention to adopt protective measures against potential ransomware attacks in the future. In the paper [12], Ami 2019 used a verification-based access control tool, and when it's used, it can stop changes to records, like renaming or canceling them. The negative points are that the clients may grant access to a task that is contaminated with infected code, and 6 proposes a method to impede malware distribution. The 2019 study also modified document and envelope access control stages to negate the advantages gained by composing access control lists. The cons are the framework in the future for experiencing the ill effects of a great deal of sharing when the observing is done on actual drives rather than fake drives.

Reference [53] present a file-based ransomware detector, Minerva, in their arXiv preprint. The paper discusses the challenges of detecting file-based ransomware attacks and proposes a solution based on a combination of static and dynamic analysis techniques. The authors demonstrate the effectiveness of Minerva in detecting ransomware attacks and suggest potential future improvements [52]. The paper [51] examines the effectiveness of machine learning models in predicting bankruptcy among Indian firms using financial data from a sample of 400 firms over a period of 10 years. The study found that machine learning models, particularly the random forest model, are effective in predicting bankruptcy. The authors also identified key financial ratios associated with bankruptcy prediction.

Reference [54] focus on the challenges and directions of detecting intrusions in the cyber-physical systems of smart cities. They discuss the unique features and complexities of smart city systems and the potential vulnerabilities that can be exploited by attackers. The authors suggest a number of future research directions to enhance the security of smart city systems and mitigate the risks of cyber-attacks. Similarly, in [54] they proposes a wearable device that uses machine learning algorithms to predict hand gestures based on muscle signals. They conducted experiments to evaluate the device's accuracy and found it to be effective for recognizing a variety of hand gestures. furthermore, The research [55] proposes a data safeguarding approach to protect against ransomware attacks that seamlessly integrates with existing systems. They evaluated their approach through experiments and found it to be effective in preventing data loss from ransomware attacks.

A. AUTOMATION TOOLS

In the realm of test automation, automation tools undoubtedly have a prominent role. Table 1 illustrates a concise summary of the most significant test automation tools and frameworks, many of which can assist companies in staying Keep abreast of the latest software testing developments. Both open-source and commercial test automation technologies are included in the list.

TABLE 1. Ransomware attacks exploit analysis.

Product	Selenium	Katalon Studio	Unified Functional Testing	Test Complete	Watir
Available since	2004	2015	1998	199	2008
Application under test	Web apps	Web (UI, API) Mobile apps	Web (UI, API) Mobile-apps, Packaged-apps, desktop	Web (UI, API) Mobile-apps, desktop apps	Web apps
Pricing	Free	Free	Dolars	Dolars	Free
Supported platform	Windows, Linux, OS X	Windows, Linux, OS X	Windows	Windows	Windows, Linux, OS X
Scripting languages	Java, C, Pearl, Python, JavaScript, Ruby, PHP	Java/Groovy	VBScript	Delphi, VBScript, Python, JavaScript, C++, PHP, C	Ruby
Programing skills	Advanced skills needed to integrate various tools	Not required recommended for advanced test scripts	Not required recommended for advanced test scripts	Not required recommended for advanced test scripts	Advanced skills needed to integrate various tools
Ease of installation and use	Require advance skills to install and use	Easy to setup and use	Complex in installation, training needed to use the tool	Easy to setup, training needed to use the tool	Advanced skills needed to integrate various tools

TABLE 2. Ransomware attacks exploit analysis.

Antiviruses	Malware Scan on request	Malware Scan On-Access	Rating of the website	Malicious URL Barriers	Protection of phishing	Detection of behavior	Vulnerability Scan	Firewall
McAfee	✓	✓	✓	✓	✓	✓	✓	✓
Webroot	✓	✓	✓	✓	✓	✓	×	✓
Bitdefender	✓	✓	✓	✓	✓	✓	✓	×
F-Secure	✓	✓	×	✓	×	✓	×	×
TrendMicro	✓	✓	✓	✓	✓	✓	×	×
Sophos Home	✓	✓	×	✓	✓	✓	×	×
Malware Bytes	✓	✓	×	✓	✓	✓	×	×
Eset	✓	✓	×	✓	✓	✓	×	×
Gdata	✓	✓	×	✓	✓	✓	×	×
Emsisoft	✓	✓	×	✓	✓	✓	×	×
Kaspersky	✓	✓	✓	✓	✓	✓	✓	×

Moreover, Umara Urooj and Mohd illustrated the detection method of malware and introduced a technique called pre-encryption crypto ransomware. It deals with pre-encryption. This concept will control the limitations of already available solutions by precisely defining the pre-encrypt boundary definition while dealing with a flexible environment [7]. This study is up-to-date and provides a brief overview of cybercrooks' attack behaviors. The author's name is Zimba, and he demonstrated in his paper the categorization of ransomware as dependent on information erasure and document encryption assault structures. From a specialist's point of view, this structure assists with revealing ransomware configuration imperfections and misusing them in information recuperation, utilizing framework volume shadow duplicates or outsider programming without paying the payoff. There are a few malicious programs that have an unalterable effect. instance, no real This arrangement could be utilized to unscramble the encoded hard circles by Syamsundar et al. [8], 2020 contributed a procedure called "detected crypto-ransomware," utilizing a grouping model from an expert's viewpoint. This paper exhibits the constraints of mark-based discovery

techniques and underscores the conduct-based recognition component's ability to distinguish crypto-ransomware. Cons are Misclassification that may occur because of choice limit blunders.

At long last, Chishimba and Zimba, who have been demonstrating ransomware battles in professional settings for the past several months, finally found success in their efforts. If available, disconnected reinforcement is provided; the ransomware infection will be mitigated. It is nearly impossible to respond quickly enough to a zero-day attack, which can break the framework. They demonstrated the associate examination of straightforward malware in terms of "why the malicious program requires a mortal touch" [9] by emailing an area bar-related to the mixing of inducement full field and query own point, thus detecting, and perceiving additional mortal angles of the cyber-attack. To eliminate issues such as alternative social-engineering approaches that are also exhaustively calculated, malware analysis paths should be completely changed in order. It should concentrate primarily on hindrance and reinstalling and saving policy in those areas associated with socio-technical results that may control and perceive client realization, data flow, behaviors,

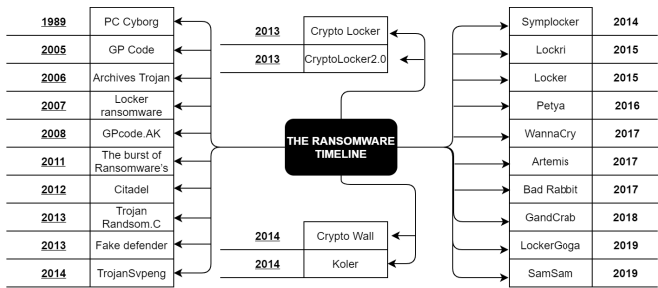


FIGURE 4. Taxonomy of ransomware timeline.

and wishes. The writer “Ana Ferreira” concludes their results as “principally detection results”.

The previous will cover socio-high-tech conclusions that might participate in managing and perceiving client knowledge, dataflow, attitude, and wishes and doubtless face ransom-attack at “their own game”- In this research [10], the authors provide a brief overview of the new inspection in the anticipation of malware assaults, and the accepted procedures to relieve the assault sway are introduced. The presence of ransomware as assistance, which works with acquiring ransom programs without any problem. This significantly reduces the section obstruction of malicious code profitable occupation, the exercises are simply anticipating that they are on the rise, clients should prepare themselves against similar assaults. Turning around ransomware encryption is very troublesome and devours time and assets. Utilizing strategies like figuring it out and cryptanalysis will contribute significantly to the decline of ransomware attacks. Methods to prevent ransomware and secure gadgets are fundamental. Ransomware developers will quickly adapt to the most recent detection apparatuses, and their advanced knowledge of various techniques will expand. An exploit analysis of ransomware attacks is shown in table 2.

B. RANSOMWARE TIMELINE

The taxonomy of ransomware timelines is shown in figure 4. The first malicious program, known as the “Cyborg virus,” was executed via floppy disc in 1889. That program was manufactured by “Joseph Popp,” a biologist, who sent out twenty thousand contaminated discs to attendees of the AIDS conference, which is the World Health Organization. AIDS information in the form of a label on the “initial floppy disks” and leaflets is involved with alerts such as “Adversely affect other program applications.” You may owe money, your computer may be damaged, and your digital machine will frequently stop working. Table 3 demonstrates a general summary of ransomware that addressed the problem, provides a proposed solution, and shows the results as well as limitations. In 2006, the virus named GP Code was released via phishing scams. It was spread through emails. Although the first one is broken easily, it creates the locked data in another location and removes other files as well. This enables the flow recovery of several files. Also, provide enough

statistics to decrypt other directories. G-Code ransom or other alternatives use symmetric encryption methods. Because of this technique, key recovery is very easy. Crypto-Locker appeared in 2013 and injects itself into the system through emails and different sites. This malicious program uses AES-256 and RSA-2048 for encrypting the files that are used in command and control and Tor networks [11]. The Petya virus was popular in the year 2016. Attackers chose the lower part of the memory configuration of the PC by removing the master-level boot record with a malicious program and then encrypting the given table. This code did not use any kind of privilege exculpation method, which is why Windows would prepare the client to get administrator rights. The same advanced encryption methods are used in Jigsaw as are used by other malware. The cyber attack creates negative threats. To encourage money, the first ransom removed data files during the attack, and every time Jigsaw was restarted, it would remove 1,000 files. In 2013, Kaspersky-sponsored Trojans Peng attempted to steal money-card information from a Russian bank. Although the primary victim was mobile phones, at the start of the next year, it was planted as a ransom for mobile phones by accusing customers of complicity. In the year 2014, a new semblance of malware evolved that targeted American mortals by delivering a negative FBI SMS and demanding 200 dollars [12]. In the year 2017, the dangerous malware Wannacry infected almost 150 nations by targeting 300,000 machines that were using the Microsoft operating system. The worm spread rapidly across several systems’ networks. It became very famous because of its infections and demanded thousands of dollars.

Reference [17] Marcus Hutchins is the one who stopped this malware. Later, the Americans found out that North Korea was behind the creation of this virus. Gandcrab was manufactured in 2018. It traveled through phishing links like emails and infected approximately forty-eight thousand nodes in a single month. Once the system is infected, the virus encrypts the files and demands payment. Then it provides a decrypted key. LockerGoga was made at the start of 2019, and the main motive of this ransom is to attack firms and industries. Files or data on the PC, the client, or the server are encrypted. The first victim was a firm called AET in France. Later, they hacked NH. Although this incident was bigger than any cybe rattack because it escalated the whole professional organization’s demands for the segregation of computer things from the network, The ransomware “SamSam” was developed in 2019 with the primary goal of targeting medical centers, such as healthcare organizations. It spies for a long time after its initial stage without being detected. The cyber attack was held in different cities in the United States, and adversaries got a benefit of approximately 6 million dollars. Kaseya announced the breach on July 2, 2021. Kaseya’s IT solutions for other organizations have had a global impact on approximately 1500 businesses, making it an excellent target. Reuters reported that the ransom demands varied from a few thousand to several million dollars. Table 4 contains an analysis of

TABLE 3. Comparison of existing literature.

Paper	Results	Problem	Proposed model	Limitation
[28]	Achieved effective ransomware detection on edge computing devices using an autoencoder classifier.	Detection of Malware is quite difficult.	Deep learning approach	- Requires further validation on diverse real-world datasets and edge environments.
[27]	Effective detection of crypto-ransomware demonstrated.	Attackers encrypt the victim's files with crypto ransomware which is nearly impossible to break.	Utilizes behavioral analysis to enhance detection accuracy	Limited discussion on real-world deployment challenges and effectiveness.
[26]	Identifies key factors contributing to ransomware success.	Malware counteraction	Provide taxonomy of ransomware threats and evaluates various countermeasure strategies	The overall success rate of the proposed technique is 40%
[19]	Could be useful for identifying malicious code.	The advance in cryptography has caused the complicating cyber-attacks.	features from process Novel model	can be extended to identify other malware.
[18]	high accuracy in detecting ransomware in executable files and effective detection rates with low false positives and false negatives.	Previous methods rely on signature-based or heuristic approaches.	machine learning-based approach for ransomware detection	require extensive feature engineering and tuning to achieve optimal results.
[33]	Proposed systems demonstrated effectiveness in preventing and tracking ransomware attacks.	Existing ransomware defenses are often reactive rather than proactive and lack effective tracking.	Introduces a dual approach: a prevention system to block attacks and a traction system to trace them.	Effectiveness may vary with new ransomware variants and potential challenges in integrating with existing security infrastructure.
[35]	Provide countermeasures against crypto-ransomware and highlights various strategies for prevention, detection, and response.	The complex ecology of Crypto ransomware makes it very dangerous.	Includes prevention, detection, and response strategies and a taxonomy categorizing countermeasures by functionality and effectiveness.	Limited empirical data on real-world effectiveness of countermeasures.

ransomware alternatives as well as methods for identifying this malware. Revil demanded 70 millions in bitcoin from Kaseya, but who paid? Kaseya declined to pay the FBI and the US Cybersecurity and Infrastructure Agency. On July 21, 2021, Kaseya obtained a universal decryptor key and sent it to the impacted firms. Because of a security breach, JBS USA, a major beef supplier in the United States, announced the closure of five US facilities on May 31, 2021. Attacks in Australia and the United Kingdom were also impacted. JBS paid an 11 million ransom in bitcoin to prevent further hardship from the hacking group. The FBI implicated REvil, a well-known ransomware organization, in the attack. On May 7, 2021, a hacker organization known as Darkside used malware to obtain access to the US's largest "refined products" pipeline. The 5,500-mile Colonial Pipeline transports about 100 million gallons of oil daily. After the attack, gas prices in the US surpassed 3 for the first time in seven years. The pipeline operator said it paid the hackers 4.4 million in bitcoin. On June 7, 2021, the DOJ announced partial reimbursement. Using a bitcoin wallet's private key, US authorities tracked the move and recovered 2.3 million. A hack by Darkside, which took 150 GB of Brenntag's data and threatened to disclose it unless ransom

demands were satisfied, was discovered on April 28, 2021, by the German chemical wholesaler Brenntag. Because of its discussions with the perpetrators, Brenntag was able to negotiate the initial 7.5 million ransom down to 4.4 million, which was paid on May 11th.

C. RANSOMWARE CLASSIFICATION

The taxonomy diagram illustrates that the victims of ransomware are often divided into two groups: end-users and organizations. The target platform is another critical consideration when understanding ransomware behavior. Most of the time, it is specifically designed for a platform, such as an objective software package, because it typically leverages the system-specific libraries and functions to perform its malicious actions. Figure 5 illustrates the explained taxonomy. The authors, DANIAL and POOIA in their research explained a unique malware detection approach that combined a quick feature extraction technique with the development of an optimum dataset using a modified genetic algorithm. The approach suggested to providing an appropriate dataset to train the malware detection model, this study used genetic algorithms to carry out the development of unusual malware through crossover and mutation processes

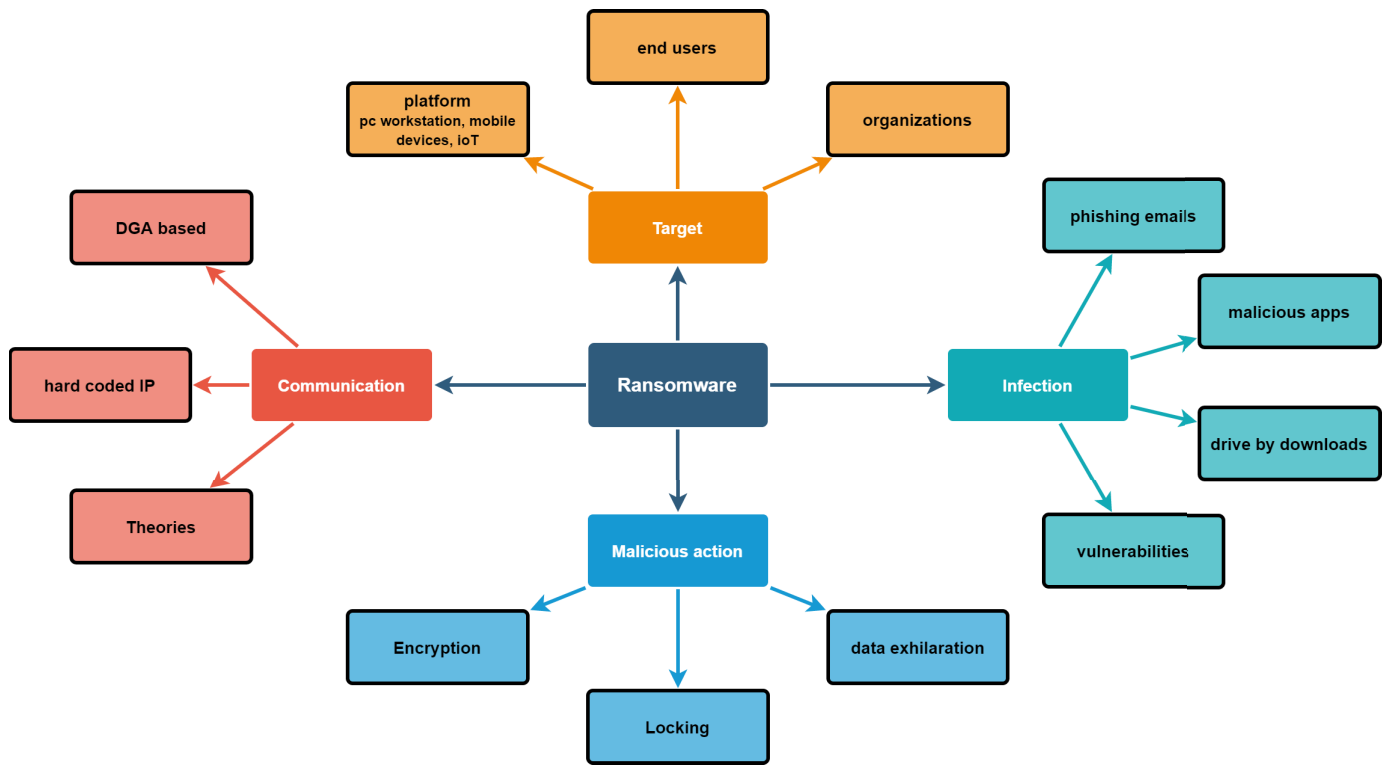


FIGURE 5. Taxonomy diagram of existing ransomware.

TABLE 4. Ransomware attacks exploit analysis.

Malware	Techniques	Encryption algorithm	Infection	Exploits	Damage potential
Crypto wall	Malicious ads Phishing links Exploit kits.	AES ₂₅₆ , RSA ₂₀₄₈	Through malicious email attachments, browser exploits.	Nuclear escapade-kit, Angler escapade-kit, Magnitude exploit kit.	High
CTB-Locker	Affected Email link, Phishing Ads, Torrent sites	RSA ₂₀₄₈	Duplicate itself to short-time data, it generates schedule function to operate restart	Rig escapade -Kit Nuclear escapade- Kit.	High
Wanna-cry	Server Message Block (SMB) Protocol vulnerability	AES ₁₂₈	Load mssecsvc.exe payload	Eternal Blue exploit, Double Pulsar exploit	High
Petya	MBR infection File encryption, User encryption	RSA ₄₀₉₆ , AES-256	Load via rundll32.exe	SMB exploits Execution across network shares	High
Crypto-Locker	Execution across network shares	RSA ₂₀₄₈	Via downloads on a secure network	Load across network shares	High
SamSam	Exploits or brute-force tactics	RSA-2048	custom infection used in targeted attacks	gang exploits vulnerabilities in an organization's Windows server	High
Teslacrypt	Torrent-sites Exploit-Kits	256 AES	Exploit duplicate itself into App-Data	Adobe flash escapade	High
Trojan Ransom	backdoor access through Rootkit, Trojan-DDoS, FakeAV	2048-bit RSA key	By sending multiple DDoS requests	contain data or code that takes advantage of a vulnerability within application software	High
Gandcrab	spam emails, exploit kits	RSA 2048-bit, AES 256-bit	Infects pc and illustrate messages demanding a fee	encrypts victims files and demands a ransom payment	High

in behavioral traits [12]. Their paper reduced classification errors for metamorphic malware and improved model accuracy. Ransomware’s primary target platforms are PCs

and workstations, mobile devices, Internet of Things (IoT), and Consumer Product Safety (CPS) devices. Ransomware authors use the infection techniques that ancient malware

authors used to infect their targets. Infection methods of ransomware are often categorized into five groups: malicious emails, SMS or instant messages, malicious applications, drive-by-download, and vulnerabilities. In the context of ransomware, the malicious actions that will be taken by ransomware are often divided into two groups: encrypting and protecting.

III. RESEARCH METHODOLOGY

The provided taxonomy represents a standardized approach to automating tests that takes advantage of various tools, integrates well, and is targeted at certain terms or patterns of behavior. Furthermore, the chart illustrates several components involved in a test automation project. This little arrow is Selenium's execution arrow, which connects the automation project and the App Under Test. In figure 6 the automation framework is shown.

A. A COMPREHENSIVE TESTING FRAMEWORK FOR AN AUTOMATION PROJECT

To create an effective test automation project, one must understand the project components and how to increase performance sustainably and efficiently. To ensure a high-quality test automation system, follow these 20 steps:

- a. Locate all page elements and configure them into a class that can be used to implement multiple tasks. For example, insert text and click a button before identifying the elements to generate a page object.
- Design a method that executes when a failure occurs.
- To establish independent tests, add another method to get the page to a particular location (e.g., the main page).
- Build a report module and implement it for the automated tests that have already been conducted.
- Make screenshots and recordings part of the test reports when they fail.
- Configure the project to enable external configuration files to define all settings, including where reports are saved.
- Use embedded visual testing for image comparison, such as moving a 6-pixel square by using picture comparison, to ensure the site appears as intended.
- Establish cross-browser compatibility.
- Create log files that log all errors and stack traces, which are helpful in the event of a failed test, providing several benefits for both developers and testers.
- Include a central location where all property values for all elements in the application can be found.
- Even if access to the application's database is unavailable, execute queries and draw data to build a portfolio simulation.
- Use the DDT technique to read CSV or Excel files.
- Use keyword-driven testing (KDT) or behavior-driven testing (BDD) to implement the Gherkin language.
- Develop parallel testing support in various settings.
- Integrate external cloud services, such as YouTube APIs.

- Schedule execution and connect to CI systems, such as Jenkins.
- Gain exposure to other mobile platforms.
- Create a desktop app version.
- Ensure the infrastructure for testing follows well-known design patterns, clean code techniques, and coding rules.
- Document the method from A to Z. Selenium is only a small part of the automated testing project on the list.

B. SELENIUM DATA DRIVEN FRAMEWORK APPROACH

In Figure 6, the successful test automation infrastructure is shown with a detailed framework layer and automated test results. In addition, Diagram 7 shows the entire Selenium data-driven framework from start to finish. The Selenium Data-Driven Framework, which utilizes Excel, is represented using the Selenium Data-Driven Framework diagram. These consist of three sections:

- Layering on the framework (which consists of core framework functions).
- The outermost layer (Test-Case-Layer) contains the application-level function (which consists of application-related functions).
- Each test case in the upper layer will keep the name "Folk" and call the configuration-level function, which in turn calls the exercise function. This setup is determined by Excel, and all test cases and test data will be included on a single Excel line, making it easier to manage. Separate browsers, such as Chrome, Mozilla, and Opera, are extremely useful. Because VSDDF is a Maven-based configuration, the client does not have to worry about dependency management. Jenkins or locally, discharge can be triggered with a single click. When any modifications to the Git code are made, Jenkins can immediately trigger the implementation. Using Test Rail APIs, VSDDF allows users to upload test results to Test Rail (a test administration tool). Logs of exercise, platform information, line bar graphs, test implementation details such as the number of fails and passes, complete test cases, each test case with line-by-line test steps, screenshots at the execution of failure, and so on are all included in the detailed report generation.

C. PURPOSED METHODOLOGY ARCHITECTURE

The Figure 8 diagram illustrates the proposed architecture of the given methodology. Firstly, the attacker sends a malicious link to the user via email or other methods, such as a document that contains the URL. Once the user clicks on that malicious link, the hidden script inside it will be executed on the victim's machine. After this, the attacker will be able to gain a session onto the victim's machine via the malicious link. Secondly, after clicking on the link, the backdoor is downloaded onto the user's public folder in C:\Users\Public\Explorer.exe, which can be used to perform further attacks. The backdoor has limited capabilities; it may receive and upload files, conduct commands, and relay the results to the attackers. Regardless of how limited the list

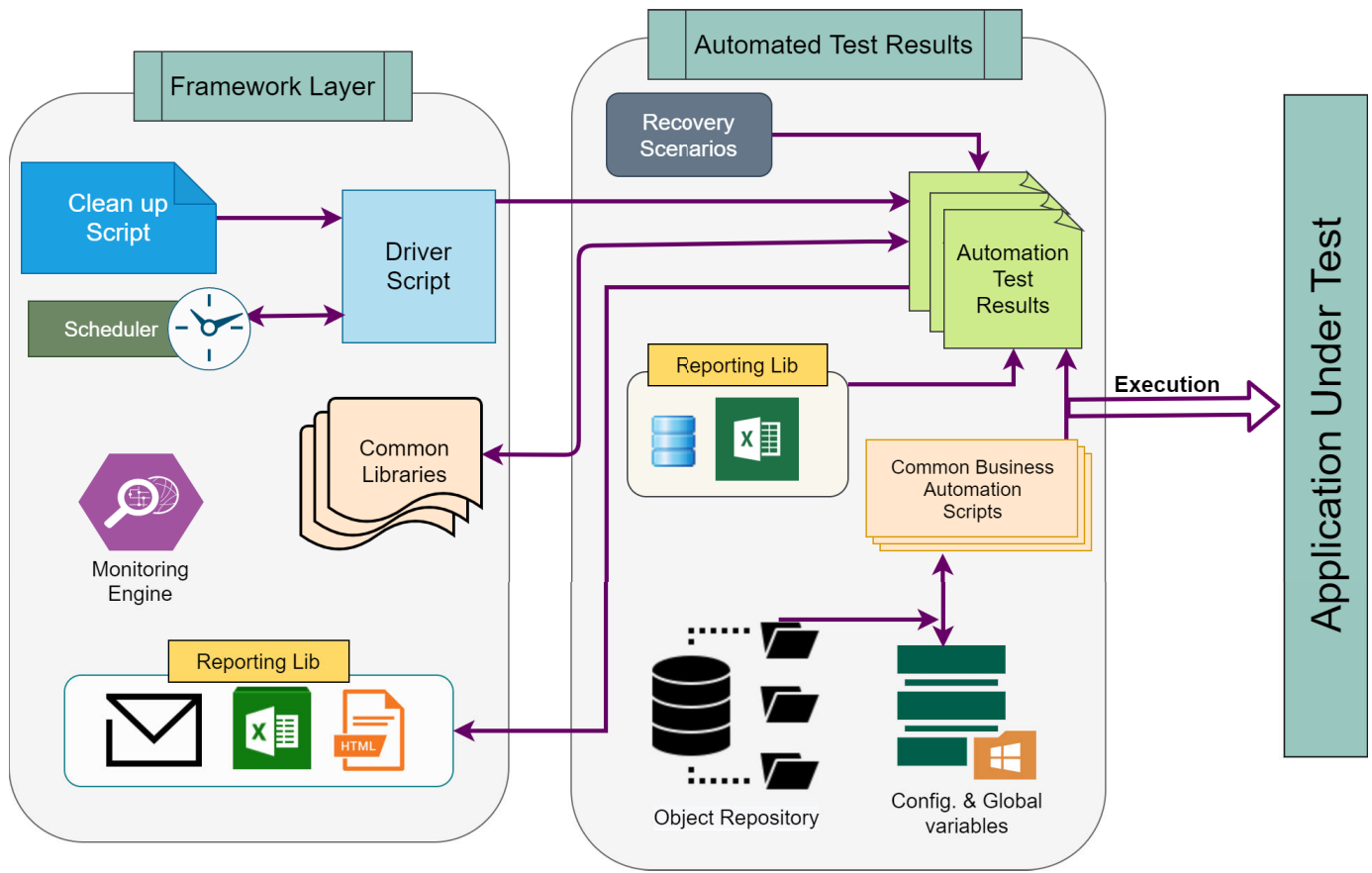


FIGURE 6. Successful test automation infrastructure.

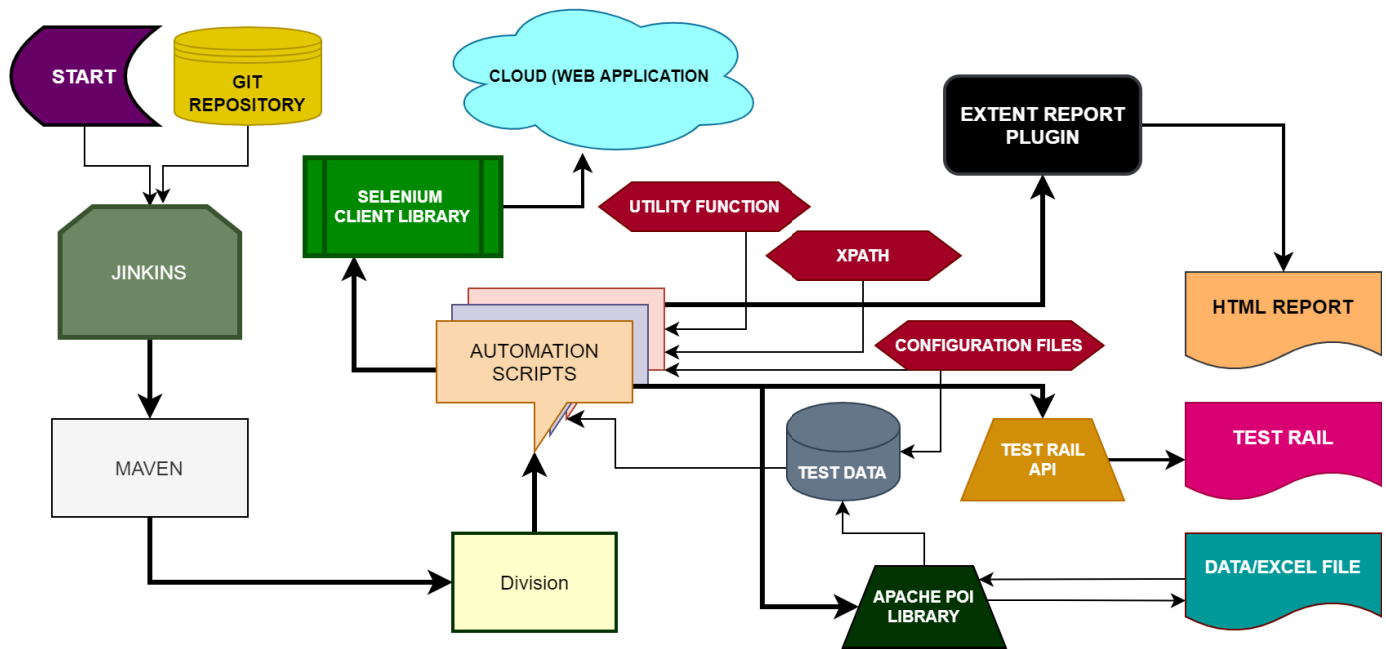


FIGURE 7. Selenium data driven framework.

is, it allows the hackers to upload and execute other tools for more reconnaissance and lateral movement. The malware exploits a standard obfuscation method known as “txt2png,”

“XorStringsNET,” and “stack strings” to generate large char strings to disguise harmful functions like persistence and CC communication from static detection.

Furthermore, the browser is launched automatically with the assistance of OneDrive, and the URL is pasted on Social-Site.com to search by navigating the finding element such as ID Xpath. Second, the social site is now logged in thanks to the tab pressing. We will find the “create post” option, and our link will be posted. Thirdly, whenever the other user on the victim’s friend list clicks that link, it will automatically download their machines in the form of a WinRAR file and open the ransomware programs executed on their operating systems, regardless of the actual action performed. For instance, with the help of malware, we can encrypt the user files, change their extensions, or even delete them. It is up to the ransomware behavior. To better evaluate our proposed approach, we mapped the collected TTPs and payload onto the MITRE ATTCK framework.

1) FULL APT ATTACK

The cyber attacks of today are sophisticated, strategic, flexible, and automated. An attack can be launched by an individual looking for information to exploit. So, in this paragraph, we demonstrate the process and staged nature of the Apt attack. In the initial stage, the attacker obtains access to the target network through APT. This is accomplished using a phishing email, a malicious attachment, or an application vulnerability. The attacker’s purpose is to use this access to infiltrate the network with malware. At this point, the network has been compromised but not breached. An infected exe has malicious code that will do an attack, which is hosted on the server.

The process of downloading and running “a.exe” in the public folder by using the technique of remote template injection “a.exe” should provide persistence by using PowerShell and scheduling task processes. Following the acquisition of persistence, we will now download and run “mal.exe” as proof of concept. The inserted virus scans for network weaknesses. It talks with external command-and-control (CC) servers to obtain instructions on how to attack these vulnerabilities and receive further malware. The algorithm explained the malicious code execution.

D. LATERAL MOVEMENT THROUGH SELENIUM

Selenium is undoubtedly the king of the web, with over 80 percent of the worldwide respondents surveyed choosing Selenium as their preferred functional testing tool. It is a tool set that automates web app testing across platforms and can simulate user interactions in a browser. In other words, the Selenium web driver is a piece of software that controls the browser by programming it to be more flexible and powerful. It supports multiple browsers on multiple platforms, such as Google Chrome. For phones and tablets, you can use Firefox 3.0+, Internet Explorer 6+, or Opera 11.5+, which were created to provide a simpler and more uniform programming interface than Java, C, Python, Ruby, PHP, and Perl.

The same Web Driver script can run on different platforms. It is an efficient Web-Driver that leverages each browser’s native support for automation.

Algorithm 1 Create Malicious Document

Input: ES (meterpreter shell code), C2 (Lhost, Lport)
OM(simple obfuscation), c-sharp (payload)
Collection: Mal-docx [], Mal-pay [], Mal-URL [], Mal-code [], C (payload)

1. **Initialize.**
2. **OM** (simple obfuscation), L-Host, MP=Payload
3. **Output.**
4. Compiled code for Macro 4.0 Excel, Payload “We use Macro 4.0”
5. **Excel-donut** (Malicious Payload)
6. **If** (MP =Macro 4.0) “Malicious code is Macro 4.0”
7. Result $\leftarrow$ call OM(MP)
8. **If**(MP=OM) “Malicious code is obfuscated”
9. Result $\leftarrow$ Macro 4.0
10. **Compiled.**(Excel 4.0 Macro) “Execution”
11. **Generate**(4.0 Macro)
12. **Embedded**(Macro 4.0, Xls)
13. **If** (MP=URL)
14. Result $\leftarrow$ generate(Mal-docx.docx) “Document generation”
15. **If** (MP=C[payload]) “c-sharp code”
16. Result $\leftarrow$ generate(Excel-xls) “Generation of malicious Xls file”
17. **If** (MP=Es)
18. Result $\leftarrow$ call(C2) “Command and control session”
19. **Compile** $\leftarrow$ MP

1) EXPERIMENT-1

In our scenario, we have already done the web automation part using C code in Visual Studio. So, first, we go to a page, then locate an element and do something with that element, and assert the result. The given code is an example of a real environment that explains how to automate the element using Google. Once the Selenium test script is finalized, we will run the script, and that Google Chrome browser test script will launch and automatically run all the test scenarios.

2) EXPERIMENT-2

The given code illustrates how we can log in to the web page in C. Now, after logging into the Facebook page, we can post our malicious link by pressing the tab key once we’re in the create post-bar. Once the link is pasted, every user who is on the friend’s list will click the link, and the actions performed on their systems will vary. For instance, the ransomware that is placed inside the link will be downloaded on their machines, and it depends on malware behavior.

In our scenario, the malware inside the link is also divided into multiple parts. Firstly, it can change the file extensions on the machine. Secondly, the middle type of malware can modify the data files and change them into corrupt forms. Thirdly, the documents are locked, and the message reads, “Your data is encrypted, please enter the keys to decrypt the data.” The machine shown on the victim’s screen in

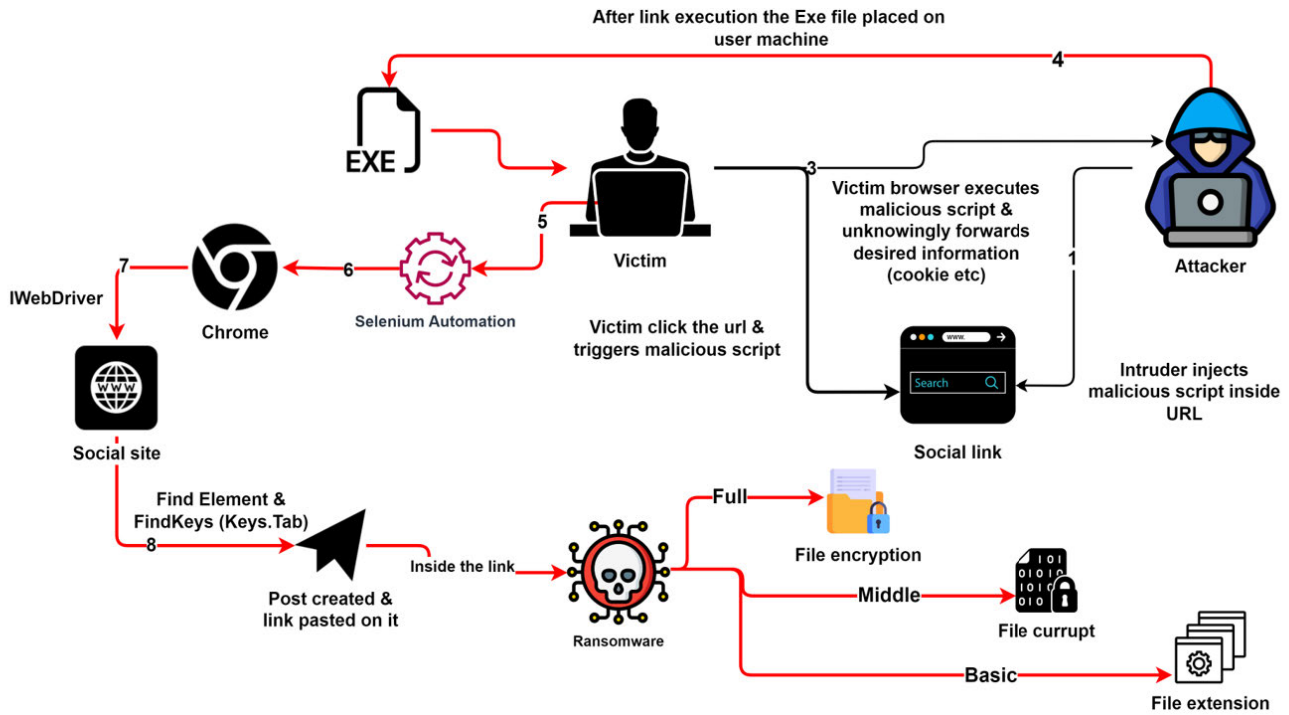


FIGURE 8. Ransomware lateral movement through selenium web automation.

Figure 9 shows the flow of our experiments. The infection chain includes spear-phishing emails that contain a file with a URL that links to an encrypted RAR file, which then installs malware. The next section demonstrates the results of our proposed methodology in the form of some diagrams.

IV. RESULTS AND DISCUSSION

According to the research questions, how is this topic unique from previous ones? Simply put, we live in a modern era where attackers use multiple tactics to access victims' machines. Every day, we hear news about companies being hacked by new ransomware and being demanded money. Our main motive for conducting this research is to reveal novel strategies that attackers may use in the future, given that most companies adopt automation tools for their work. Our approach aims to provide a better understanding of new malware-spreading techniques that attackers may use in the future. The main purpose of this approach is to gain knowledge about how attackers think and how we can defend our systems. Our experiments are divided into several parts. First, we implement the ransomware encryption technique, which is shown in the given diagrams. Then, we show the web automation results in diagrams. Figure 10 displays the Code Execution First stage of the ransomware, and figure 11 illustrates the results after malware execution.

A. EXPERIMENTAL RESULTS 1

We conducted our experiments in Visual Studio using C-Sharp code, as previously discussed. Our implementation is divided into several parts. Firstly, the picture demonstrates that the file extensions are changed after clicking the

ransomware program file. Similarly, further DES encryption is performed, and the results are presented with better explanations. The given code demonstrates the file extension parts that are converted into another format. However, the screenshots in Figure 10 illustrate that after clicking the malicious .exe program in the Windows environment, the files are converted into other formats like encryptMe-0.crypt, and the malware notification also displays the operating system name, IP address, and MAC address to show the user that their system is being hacked by someone. Every time the victim runs the downloaded .exe, their systems are hacked, and their files are encrypted by the malware.

As we can see, our experiments demonstrate the first file results, which illustrate that the EncryptMe-0.txt file is encrypted, and the original file is shown with its actual data formats. Further pseudo-code is also illustrated in the following images, such as encrypted files, encryption codes, and encrypted formats. These are the full forms of ransomware that are executed.

B. EXPERIMENTAL RESULTS 2

An ideal solution for those looking to improve website load times and navigate quickly may be found in the Selenium web automation results, which are provided in detailed graphs that correlate the website load generated with navigation timing data related to all web pages or a specific page.

C. EXPERIMENTAL RESULTS 2

We conducted trials and concluded that if we execute the program, we have around an 85 percent chance of success against Chrome, Safari, and Firefox. We conclude from

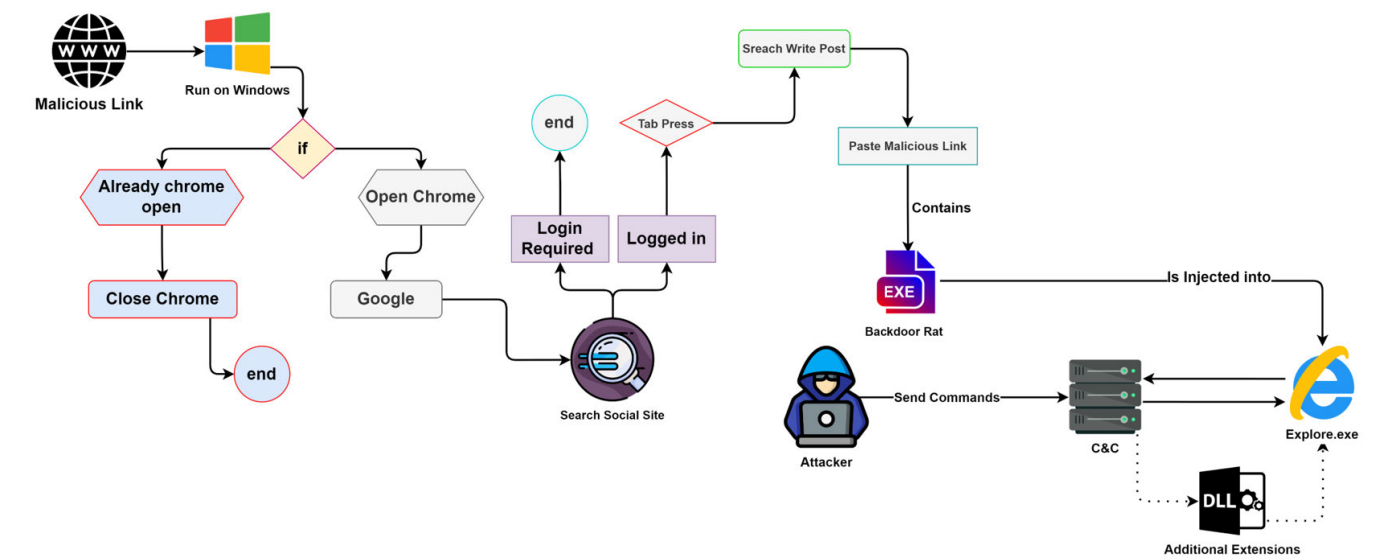


FIGURE 9. Flow diagram of the Experiments.

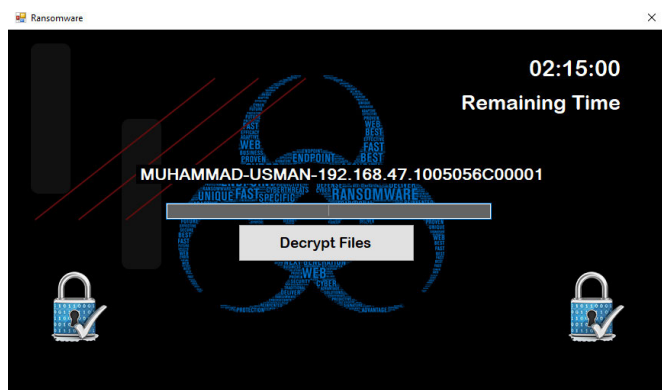


FIGURE 10. First stage of the code execution.

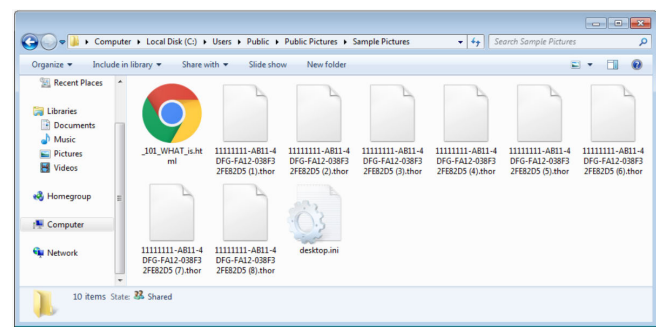


FIGURE 11. Results of the malware after execution.

the results that Firefox is more vulnerable compared to other web browsers. However, in the real world, the actual browsers operate in front of users to accomplish tasks (the user interface). The user codes and executes the script, and a browser is launched and does what it is instructed to do. Real browsers are Google Chrome, Firefox, and Internet Explorer. The success of the attack depends on multiple things listed

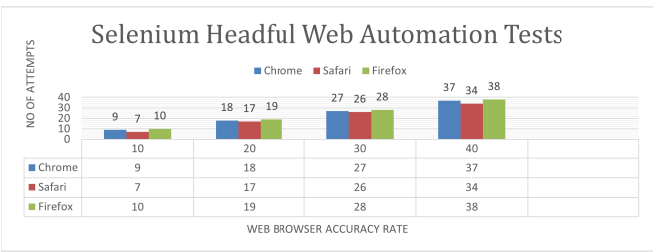


FIGURE 12. Headful web automation tests.

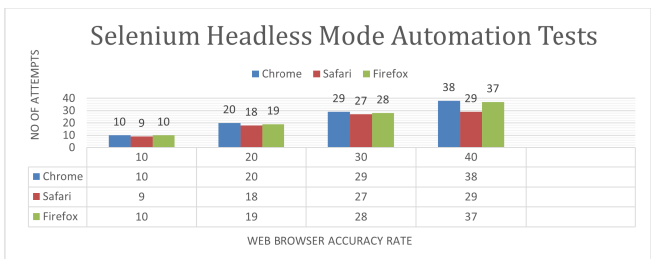


FIGURE 13. Headless web automation tests.

below: • System specifications, Operating system • Internet speed • Web server bandwidth • Browser • Automation tool • Anti-malware

The strategy discussed in our paper is successful on our operating system, and the success rate is 85 percent. Every time we execute a malicious program on multiple systems, we get accurate results based on several points that we have discussed before. By performing experiments on different browsers, we kept all the parameters constant, such as internet speed, the same Windows operating system, web server bandwidth, etc. Figure 12 demonstrates the selenium web test results. The main problem arises when the victim’s operating system displays all the automation

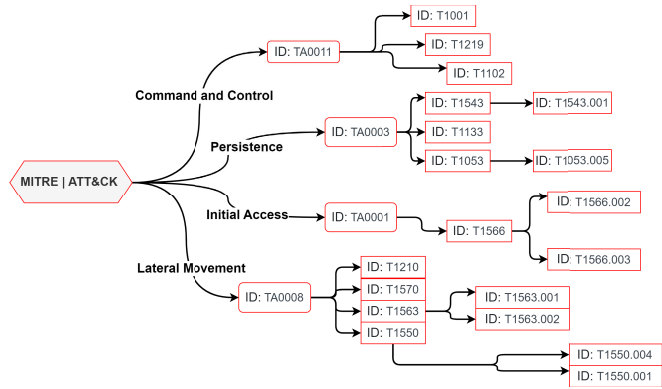


FIGURE 14. MITRE ATT&CK mapping.

on the screen, which can be seen by the victim. This issue specifically occurs in Google Chrome. While Selenium supports headless logged-in profiles in Firefox, Opera, and Edge web automation, it does not support headless logged-in profiles in Chrome due to Chrome's restrictions. Instead, we use a different approach where we open a profiled Chrome, resize it to the dimensions of 10px, and move the window away from the screen. In a headless environment, back-end instructions and functions are executed without the user interface (UI). During web automation, users write and run code without a browser appearing, while back-end testing is performed. This method can improve the accuracy rate of headless web automation, as demonstrated in Figure 13, which shows the selenium head web test results. For instance, in the first attempt, the Chrome windows run the Selenium script in 2 seconds on a headful scenario, but when we use headless web automation, the accuracy rate improves by 1 second. After testing, we found that we have a 95 percent chance of success against Chrome, Safari, and Firefox if we execute the script headless. We also inferred from the data that Firefox is more susceptible compared to other web browsers.

Furthermore, we mapped our techniques onto MITRE, and 14 figures in total illustrate the methods used in this paper.

V. FUTURE TRENDS

We predict that the number of malware attacks in 2023 will be 60 percent higher than in 2021, using novel strategies. Therefore, the future research direction should focus more on malware protection and finding new ways and algorithms to fight against malware. From a protective perspective, several kinds of questions come to mind. What is formal malware protection? Which antivirus program should you install? A broad range of services is available, and independent antivirus testing labs frequently obtain perfect or nearly perfect ratings for Bitdefender and Kaspersky. One McAfee Antivirus subscription assists in installing security for all Windows devices, Android, Mac OS, and iOS. Webroot Secure Anywhere Antivirus is the least expensive antivirus program available due to its new behavioral technology. We named these four antivirus options for editors. However, pricing is

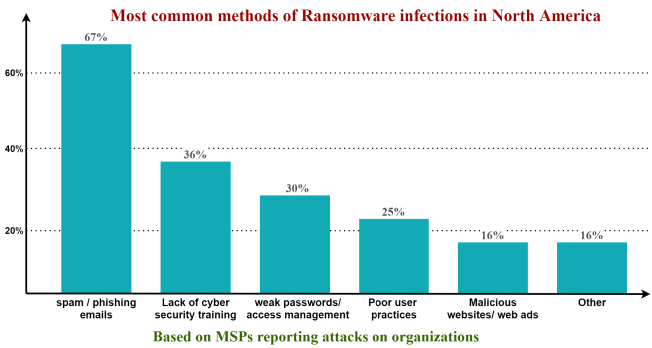


FIGURE 15. Ransomware infection types.

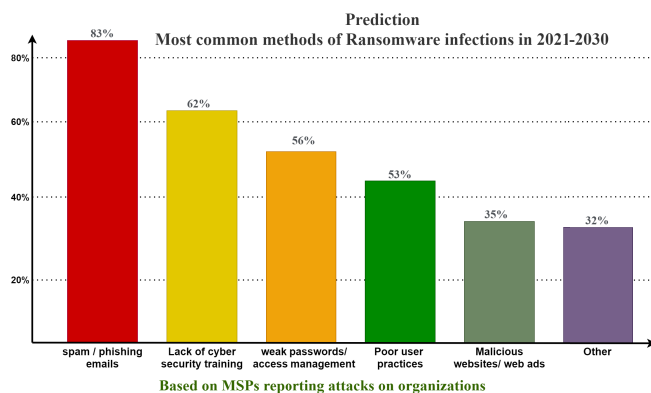
not the only factor to consider when selecting a solution. Analyze the reviews for our highest-rated goods and build your case similarly. The table below summarizes the best antivirus programs for detecting and preventing malware. Figure 15 demonstrates that scams and phishing emails will be the most dangerous and common forms of threats used by cyber crooks in the coming years. Figure 16 shows predictions for ransomware infections from 2021 to 2030. On Friday, 2/6/2021, around 2 p.m. ET, an extremely sophisticated ransomware attack hit hundreds of US firms using management software from a Florida-based supplier called Kaseya. The attackers modified a tool called VSA, which was used by smaller firms that handle technology. They then simultaneously encrypted the files of the clients of those providers. Huntress, a security firm, tracked eight managed service providers infecting 200 customers. Kaseya claimed it is researching a "possible assault" on VSA, which IT professionals use to manage servers, desktops, network equipment, and printers. Based on MSPs' reports on cyber threats in 2021, we predict several methods of infection will increase between 2021 and 2030. We assume that since the world will be entirely dependent on digital technology in the coming years, adversaries will perform new actions to spread their ransomware. The attacker adopted several novel tactics that can be improved upon in the future.

A. OPEN RESEARCH ISSUES AND CHALLENGES

Cyber attacks have rapidly increased in the last year due to the shift towards remote work. Ransom attacks, which increased by 150 percent from the previous year, were on the rise in 2020. The amount paid out by victims of terrorist crimes is expected to rise by 300 percent in 2020. This trend began in early 2021 and is continuing. Malicious software (malware) attacks on private enterprises, critical infrastructure, and municipalities have received considerable attention in recent months. Additionally, ransom demands have dramatically escalated in 2021, with some exceeding tens of millions of dollars. Furthermore, attacks have become more sophisticated, with threat actors capturing key firm data and holding it hostage in exchange for cash. Similarly, adversaries are adopting new techniques for hiding malware, which are powerful enough to evade detection

TABLE 5. Bot detection techniques:

Behavioral Detection	• Mouse cursor movements (randomized vs linear, on-linear, patterned) • Clicks of the mouse (bots might have certain uniformed rhythms). • With the key pressed, scroll. • The maximum number of requests that can be made during a session • The total number of pages viewed in a session. • The existence of a pattern and the sequence in which pages are viewed. • The average amount of time spent between pages
Fingerprinting Detection Technique	• 1. Browser fingerprinting • 2. Checking browser consistency • 3. Checking operating system consistency (user agent). • 4. Inconsistent behavior • 5. determining whether the browser is operating in a virtual environment.
Using CAPTCHA	• audio and image recognition are popular in CAPTCHA applications.
End Words	• You need powerful anti-bot software to protect your computer from things like DDoS, identity theft, data/content scraping, and more.

**FIGURE 16. Prediction on ransomware infections from 2021 to 2030.**

even by modern anti-viruses. Notable attacks on well-known manufacturers have already occurred in June. This report in Figure 19 shows that ransomware groups accounted for all 292 firms' cybersecurity breaches. The criminals involved in these heists have raked in nearly 45 million so far by holding their victims as hostages. The Kaspersky team has divided the most dangerous types of ransomware into five categories. The most active groups target companies, encrypting data and demanding a ransom. They published a report in 2022 and stated that these were the most harmful programs: 1. Maze (aka ChaCha ransomware), 2. Conti (aka IOCP ransomware), 3. REvil (aka Sodin, Sodinokibi ransomware), 4. Netwalker (aka Mailto ransomware), and 5. DoppelPaymer ransomware. Even though it is widely accepted that emails are the primary way to infect computer

In order to test your bot with a real world problem, your bot has to [solve the challenge below](#).

Your Behavioral Score: **0**

systems with malware, some individuals still fall prey to malevolent social engineering, and as a result, entire systems are compromised with potentially destructive ransomware. Ransomware infections often result from user or admin errors when it comes to basic cybersecurity practices, such as the use of weak passwords, a lack of password access management, and a lack of user awareness. The diagram illustrates several different types of malware infections that cyber criminals are likely to use in 2021 in North America.

VI. PROPOSED SOLUTION AND DISCUSSION

In this section, we present a comprehensive method for detecting and preventing web automation, specifically focusing on Headless Chrome Detection. Several points are used for bot detection in Chrome, such as behavioral classification, additional sources and links, bot challenges, new and old detection tests, HTTP headers, TCP/IP fingerprints, TLS fingerprints, browser fingerprints, canvas fingerprints, WebGL fingerprints, web workers, service workers, cross-pings, and browser data. The "behavioral classification score" provides a rating between 0 (for bots) and 1 (for humans) based on advanced behavioral heuristics. A score below 0.5 indicates that you are most likely a bot. The score is computed after seven seconds of browsing the site. If you want to obtain a new score, refresh this page.

- We use a novel technique that cyber attackers might adopt in the future.
- Using a selenium web automation tool is the best way to let ransomware move laterally on social networks. We got this idea from the existing CVE-2022-21882, CVE-2020-0796, and CVE-2021-35211.
- The proposed approach has been applied with 95 percent success on our Windows operating systems.
- Our application has been tested on a variety of platforms, including Google Chrome, Firefox, and Safari.
- Our research shows that there are several ways to spread ransomware.

VII. CONCLUSION AND FUTURE RESEARCH DIRECTIONS

Ransomware has long been the most dangerous and costly type of malware infection in the cyber world. While email remains the most common method of infection, URLs are also a significant entry point. Data loss resulting from ransomware attacks can have a severe impact on both organizations and enterprises. To protect your system from ransomware, it is essential to learn what it is, how it infects a computer, and what to do if an attack occurs. However, the most basic technique to prevent any type of malicious software is to use up-to-date antivirus software.

In our research paper, we propose a novel methodology that cyber attackers might adopt in the future. Our recommended technique is to enable ransomware lateral movement on social networks using a Selenium web automation tool. We derived this concept from existing open-source malware programs, including CVE-2022-21882, CVE-2020-0796, CVE-2021-35211, and Lime Rat, GonnaCry, Crypter, and PSRansom, which are available on GitHub and can access any PC (victim's IP address) through a link constructed on the Paste Bin site. When the user clicks the link, the virus becomes active on the machine, allowing the attacker to penetrate the user's PC by adding the EXE file. The ransomware is then executed in a real environment using Selenium automation.

We chose a social site for our implementation. When another user sees the post, the malicious link will appear on the social site, directing the Google window and the social app to open, and requiring the user to enter their login credentials again. By providing their information, they inadvertently allow us to break into their devices. The malicious link will also appear on the screens of their social account friends. When the user bot clicks the upload button, access to their details is granted, and a ransomware application is downloaded to their computers via the URL, which can encrypt their data files.

Our proposed approach has been successfully applied with 95 percent accuracy on our Windows operating systems. However, according to research conducted in 2022, numerous technologies, such as Google Bot, can detect malicious web executions based on behavioral and fingerprinting detection techniques. We also show numerous new versions of antivirus software in a specific column to determine which antivirus software provides the best defense against ransomware while also meeting user demands.

In conclusion, this article delves into the details of web automation technologies that hackers may use in the future to take total command and control sessions of users' machines. It is crucial to stay vigilant and protect your systems against ransomware attacks by keeping your antivirus software up-to-date and being aware of the potential risks associated with URLs and social networks.

Future studies on automated ransomware attacks may focus on three key areas in order to improve defenses and reduce threats. One interesting possibility is the creation of sophisticated detection algorithms that use machine learning and artificial intelligence to detect and eradicate ransomware before it launches its payload. Researchers could be looking at new approaches for understanding ransomware activity patterns and how they interact with system data and network traffic. Another significant issue is to improve automated response methods that can immediately isolate and repair compromised systems, hence reducing damage and recovery time. Furthermore, it is critical to investigate enhanced ransomware prevention solutions, such as improved backup procedures and strong encryption practices. Investigating the effects of recent advances, such as quantum computing, on ransomware encryption and protection tactics may potentially provide useful information.

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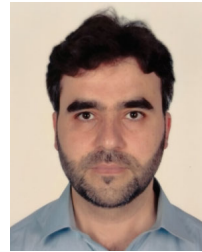
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